SUGAR TRAP ANALYSIS

INSTALL LIBRARIES NEEDED FOR THE PROJECT

```
# Install any missing libraries
!pip install pandas matplotlib seaborn plotly -q
```

IMPORTING LIBRARIES

```
# Import everything we need
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings

warnings.filterwarnings('ignore')

print("All libraries loaded successfully!")
```

```
All libraries loaded successfully!
```

DATA INGESTION & THE CLEAN UP - STORY 1 (CELLS 3-12)

```
!wget -q --show-progress "https://static.openfoodfacts.org/data/en.openfoodfacts.org.products.csv.gz" -O food_facts.

print("downloaded successfully")
```

```
food_facts.csv.gz   100%[===================>]   1.16G  34.9MB/s    in 33s
downloaded successfully
```

LOAD THE DATASET

```
df_raw = pd.read_csv(
    'food_facts.csv.gz',
    nrows=500000,
    sep='\t',               # This file uses TAB not comma as separator
    low_memory=False,       # fix data type issue, read the entire file before derterming data type
    on_bad_lines='skip'     # Skip any corrupted rows
)

print(f"Dataset loaded")
print(f"Shape: {df_raw.shape[0]:,} rows × {df_raw.shape[1]} columns")
```

```
Dataset loaded
Shape: 500,000 rows × 210 columns
```

PREVIEW THE RAW DATA

```
df_raw.head()
```

	code	url	creator	created_t	created_datetime	last_modified_t	last_modified_datetime	la
0	54	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1582569031	2020-02-24T18:30:31Z	1733085204	2024-12-01T20:33:24Z	
1	63	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1673620307	2023-01-13T14:31:47Z	1750061386	2025-06-16T08:09:46Z	
2	114	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1580066482	2020-01-26T19:21:22Z	1751035658	2025-06-27T14:47:38Z	
3	431	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1714301712	2024-04-28T10:55:12Z	1714301721	2024-04-28T10:55:21Z	
4	105	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1572117743	2019-10-26T19:22:23Z	1738073570	2025-01-28T14:12:50Z	

5 rows × 210 columns

VIEW ALL COLUMN NAMES

```
print(df_raw.columns.tolist())
```

```
['code', 'url', 'creator', 'created_t', 'created_datetime', 'last_modified_t', 'last_modified_datetime', 'last_modifi
```

PREVIEW KEY COLUMNS & CHECK MISSING VALUES

```
key_columns = [
    'product_name',
```

```
    'categories_tags',
    'sugars_100g',
    'proteins_100g',
    'fat_100g',
    'fiber_100g',
    'energy_100g',
    'ingredients_text'
]

# Show the first 10 rows of only our key columns
print("Preview of key columns:\n")
print(df_raw[key_columns].head(10))

# Count how many empty cells exist in each of our key columns
print("\nMissing values in key columns:\n")

missing = df_raw[key_columns].isnull().sum()   # Count empty cells per column
total   = len(df_raw)                           # Total number of rows loaded




# Loop through each column and print the missing count + percentage
for col in key_columns:
    pct = (missing[col] / total) * 100          # Calculate % missing
    print(f"  {col}: {missing[col]:,} missing ({pct:.1f}%)")
```

```
Preview of key columns:

                           product_name  \
0         Limonade artisanale a la rose
1                       M&amp;M white
2                         Chocolate n3
3                       Pâte de fruits
4  Paleta gran reserva - Sierra nevada-
5                                  NaN
6                                  NaN
7                                  NaN
8      Confiture extra citron de Menton
9                                  NaN

                                  categories_tags  sugars_100g  \
0                                             NaN          NaN
1                                             NaN          NaN
2                                             NaN          NaN
3                                             NaN          NaN
4  en:beverages-and-beverages-preparations,en:bev...          NaN
5                                             NaN          NaN
6                                             NaN          NaN
7                                             NaN          NaN
8                                             NaN          NaN
9                                             NaN          NaN

   proteins_100g  fat_100g  fiber_100g  energy_100g  \
0            NaN       NaN         NaN          NaN
1            NaN       NaN         NaN          NaN
2            NaN       NaN         NaN          NaN
3            NaN       NaN         NaN          NaN
4            NaN       NaN         NaN          NaN
5            NaN       NaN         NaN          NaN
6            NaN       NaN         NaN          NaN
7            NaN       NaN         NaN          NaN
8            NaN       NaN         NaN          NaN
9            NaN       NaN         NaN          NaN

                                  ingredients_text
0                                              NaN
1  Weizenmehl, Rapsöl, Speisesalz, 1,7% Meersalz,...
2                                              NaN
3                                              NaN
4  Thiamin, Biotin, Chromium, Garcinia cambogia f...
5                                              NaN
6                                              NaN
7                                              NaN
8                                              NaN
9                                              NaN

Missing values in key columns:

  product_name: 15,638 missing (3.1%)
  categories_tags: 231,130 missing (46.2%)
  sugars_100g: 398,211 missing (79.6%)
  proteins_100g: 395,203 missing (79.0%)
  fat_100g: 395,339 missing (79.1%)
  fiber_100g: 425,859 missing (85.2%)
```

SELECT KEY COLUMNS & CREATE WORKING DATAFRAME

```
key_columns = [
    'product_name',
    'categories_tags',
    'sugars_100g',
    'proteins_100g',
    'fat_100g',
    'fiber_100g',
    'energy_100g',
    'ingredients_text'
]

# Create a new DataFrame with only our key columns
df_work = df_raw[key_columns].copy()
```

```
print(f"Working DataFrame created!")
print(f"Shape: {df_work.shape[0]:,} rows × {df_work.shape[1]} columns")
```

```
Working DataFrame created!
Shape: 500,000 rows × 8 columns
```

DROP ROWS WITH MISSING CRITICAL VALUES

```
rows_before = len(df_work)                    # Save count BEFORE dropping
print(f"Rows before dropping nulls: {rows_before:,}")

df_work = df_work.dropna(subset=[
    'product_name',   # Must have a name
    'sugars_100g',    # Must have sugar data (our X axis)
    'proteins_100g'   # Must have protein data (our Y axis)
])

rows_after = len(df_work)                     # Save count AFTER dropping
print(f"Rows after dropping nulls:  {rows_after:,}")
print(f"Actual rows removed: {rows_before - rows_after:,}")
```

```
Rows before dropping nulls: 500,000
Rows after dropping nulls:  100,061
Actual rows removed: 399,939
```

FILTER OUT BIOLOGICALLY IMPOSSIBLE VALUES(OUTLIERS)

```
rows_before = len(df_work)
print(f"Rows before outlier removal: {rows_before:,}")

df_work = df_work[
    # Sugar must be between 0 and 100
    (df_work['sugars_100g']   >= 0) & (df_work['sugars_100g']   <= 100) &
    # Protein must be between 0 and 100
    (df_work['proteins_100g'] >= 0) & (df_work['proteins_100g'] <= 100) &
    # Fat must be between 0 and 100
    (df_work['fat_100g']      >= 0) & (df_work['fat_100g']      <= 100) &
    # Energy must be positive (calories can't be negative)
    (df_work['energy_100g']   >= 0)
]

rows_after = len(df_work)
print(f"Rows after outlier removal:  {rows_after:,}")
print(f"Actual outliers removed: {rows_before - rows_after:,}")
```

```
Rows before outlier removal: 100,061
Rows after outlier removal:  99,694
Actual outliers removed: 367
```

RESET INDEX & CONFIRM CLEAN DATASET

```
df_work = df_work.reset_index(drop=True)


print("Data cleaning complete\n")
print(f"Final clean dataset: {len(df_work):,} rows\n")

# Check remaining missing values in our clean dataset
print("Remaining missing values:\n")
for col in df_work.columns:
    missing = df_work[col].isnull().sum()
    pct = (missing / len(df_work)) * 100
    print(f"  {col}: {missing:,} missing ({pct:.1f}%)")

# Preview the clean data
print("\nPreview of clean data:\n")
print(df_work.head())
```

```
Data cleaning complete!

Final clean dataset: 99,694 rows

Remaining missing values:

  product_name: 0 missing (0.0%)
  categories_tags: 51,549 missing (51.7%)
  sugars_100g: 0 missing (0.0%)
  proteins_100g: 0 missing (0.0%)
  fat_100g: 0 missing (0.0%)
  fiber_100g: 27,807 missing (27.9%)
  energy_100g: 0 missing (0.0%)
  ingredients_text: 57,087 missing (57.3%)

Preview of clean data:

                 product_name            categories_tags  sugars_100g  \
0                  Pinto Bean  en:asian-style-ready-meal     4.900000
1        Croquetas de bacalao                        NaN     1.900000
2           Keto & GF Granola                        NaN     3.225806
3      Ben's Pure Maple Cream                        NaN    37.720000
4  Guimauve chocolat smarties                        NaN    65.000000

   proteins_100g  fat_100g  fiber_100g  energy_100g ingredients_text
0      17.500000  10.20000    3.800000         9.00              NaN
1       5.900000  12.10000         NaN       856.00              NaN
2      19.677419  54.83871    9.677419      2770.00              NaN
```

```
3       5.680000  30.28000         NaN      2221.96              NaN
4       5.700000  19.00000         NaN      1982.00              NaN
```

SAVE THE CLEAN DATA AS CSV

```
# save the clean DataFrame as a checkpoint
df_work.to_csv('food_facts_clean.csv', index=False)

print("Clean dataset saved as food_facts_clean.csv")
print(f"Saved {len(df_work):,} rows")
print("Note: final dashboard dataset will be saved in the last cell")
```

```
Clean dataset saved as food_facts_clean.csv!
Saved 99,694 rows
Note: final dashboard dataset will be saved in Cell 26
```

THE CATEGORY WRANGLER - STORY 2 (CELLS 13-16)

CLEAN THE CATEGORY TAG

```
# The categories_tags column looks like this:
# "en:snacks,en:sweet-snacks,en:biscuits,en:chocolate-biscuits"


df_work['categories_clean'] = (
    df_work['categories_tags']   # point at the raw category column
    .fillna('')                  # replace empty cells with empty string
    .str.lower()                 # convert all text to lowercase
    .str.replace(' ', '')        # remove all spaces
)

# create a new column called categories_clean with the cleaned tags
print("Category tags cleaned")
print("\nSample of cleaned tags:\n")
print(df_work['categories_clean'].head(10))
```

```
Category tags cleaned!

Sample of cleaned tags:

0                            en:asian-style-ready-meal
1
2
3
4
5
6    en:beverages-and-beverages-preparations,en:bev...
7
8
9
Name: categories_clean, dtype: object
```

ASSIGN PRIMARY CATEGORY TO EACH PRODUCT

```
# This function takes the cleaned tags of one product
# and returns the category it belongs to

def assign_category(tag_string):

    # check if any cookie related keywords exist in the tags
    if any(word in tag_string for word in [
        'cookie', 'biscuit', 'wafer', 'cracker', 'digestive'
    ]):
        return 'Cookies & Biscuits'

    # check if any chip related keywords exist in the tags
    elif any(word in tag_string for word in [
        'chip', 'crisp', 'popcorn', 'puff', 'pretzel', 'tortilla'
    ]):
        return 'Chips & Crisps'

    # check if any chocolate or candy related keywords exist in the tags
    elif any(word in tag_string for word in [
        'chocolate', 'candy', 'confection', 'sweet', 'gummy',
        'caramel', 'lollipop', 'marshmallow'
    ]):
        return 'Chocolate & Candy'

    # check if any protein or nutrition bar keywords exist in the tags
    elif any(word in tag_string for word in [
        'protein', 'nutrition', 'energy-bar', 'granola-bar',
        'cereal-bar', 'sport', 'fitness'
    ]):
        return 'Protein & Nutrition Bars'

    # check if any cereal or breakfast keywords exist in the tags
    elif any(word in tag_string for word in [
        'cereal', 'granola', 'oat','muesli', 'breakfast',
        'porridge', 'cornflake'
    ]):
        return 'Cereals & Breakfast'

    # check if any nut or seed keywords exist in the tags
    elif any(word in tag_string for word in [
```

```
            'nut', 'seed', 'almond', 'cashew', 'peanut',
            'pistachio', 'walnut', 'hazelnut'
        ]):
            return 'Nuts & Seeds'

        # check if any dairy related keywords exist in the tags
        elif any(word in tag_string for word in [
            'yogurt', 'yoghurt', 'cheese', 'dairy', 'milk',
            'fromage', 'cream'
        ]):
            return 'Dairy Snacks'


        else:
            return 'Other'

# run the function on every single row in our DataFrame
# results are stored in a new column called primary_category
df_work['primary_category'] = df_work['categories_clean'].apply(assign_category)

print("Categories assigned")
print(f"Total products categorized: {len(df_work):,}")
```

```
Categories assigned!
Total products categorized: 99,694
```

CHECK THE CATEGORY DISTRIBUTION

```
# count how many products are in each category
# sort from largest to smallest
category_counts = df_work['primary_category'].value_counts()

print("Products per category:\n")
for category, count in category_counts.items():
    pct = (count / len(df_work)) * 100        # calculate percentage
    bar = '#' * int(pct)                       # visual bar using # characters
    print(f"  {category:<30} {count:>6,}  ({pct:>5.1f}%)  {bar}")

print(f"\nTotal products: {len(df_work):,}")
```

```
Products per category:

  Other                          69,365  ( 69.6%)  ######################################################################
  Cereals & Breakfast            11,195  ( 11.2%)  ###########
  Dairy Snacks                    7,075  (  7.1%)  #######
  Chocolate & Candy               5,334  (  5.4%)  #####
  Cookies & Biscuits              2,827  (  2.8%)  ##
  Nuts & Seeds                    2,018  (  2.0%)  ##
  Chips & Crisps                  1,598  (  1.6%)  #
  Protein & Nutrition Bars          282  (  0.3%)

Total products: 99,694
```

PREVIEW ONE PRODUCT FROM EACH CATEGORY

```
# loop through each unique category
# print one example product from each bucket
# this confirms our bucketing logic is working correctly
for category in df_work['primary_category'].unique():

    # filter the dataframe to only show rows from this category
    example = df_work[df_work['primary_category'] == category].iloc[0]
    print(f"  {category}")
    print(f"    Product : {example['product_name']}")
    print(f"    Sugar   : {example['sugars_100g']}g")
    print(f"    Protein : {example['proteins_100g']}g")
    print()
```

```
  Other
    Product : Pinto Bean
    Sugar   : 4.9g
    Protein : 17.5g

  Cookies & Biscuits
    Product : Donut Milka
    Sugar   : 18.0g
    Protein : 6.0g

  Chocolate & Candy
    Product : Véritable pâte à tartiner noisettes chocolat noir
    Sugar   : 32.0g
    Protein : 8.0g

  Chips & Crisps
    Product : Tostadas
    Sugar   : 0.0g
    Protein : 5.0g

  Protein & Nutrition Bars
    Product : PRO-TF Chocolate
    Sugar   : 5.88235294117647g
    Protein : 58.8235294117647g

  Dairy Snacks
    Product : greek yogurt
    Sugar   : 9.3333333333333g
    Protein : 7.3333333333333g

  Cereals & Breakfast
```

```
    Product : Purée de pommes de terre
    Sugar   : 1.64g
    Protein : 2.27g

  Nuts & Seeds
    Product : pumpkin seeds
    Sugar   : 1.7g
    Protein : 35.0g
```

THE NUTRIENT MATRIX - STORY 3 (CELLS 17-20)

PREPARE DATA FOR VISUALIZATION

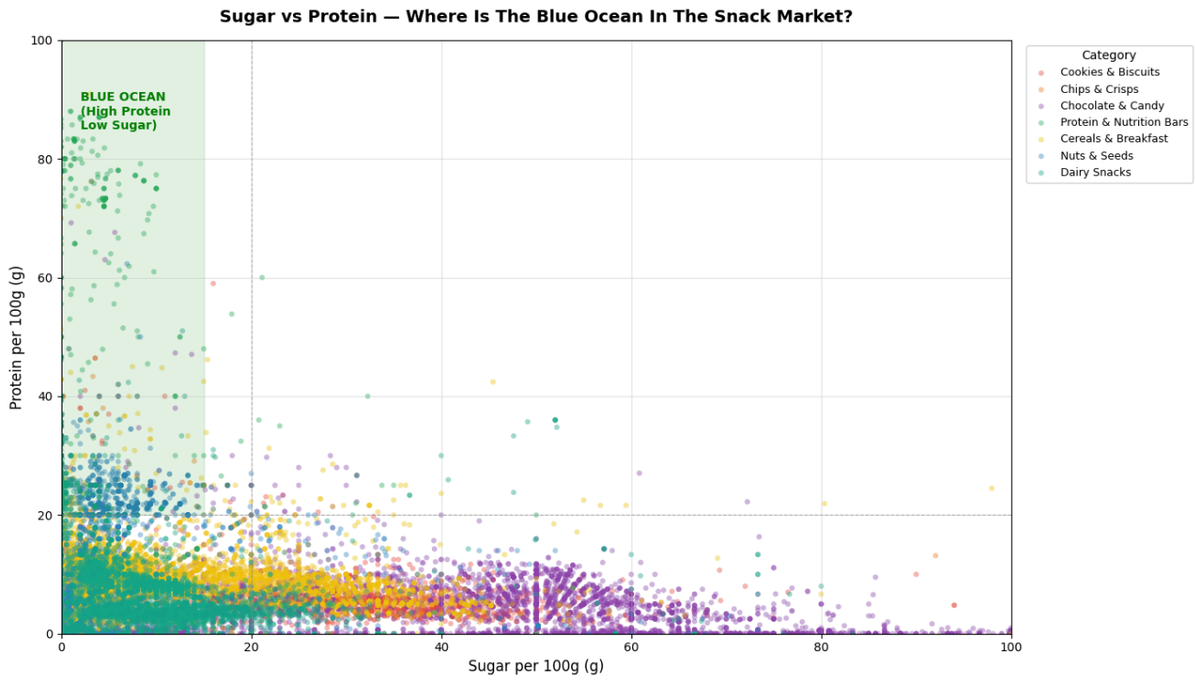
```
# we exclude the Other category from our charts
df_viz = df_work[df_work['primary_category'] != 'Other'].copy()

# reset the index after filtering
df_viz = df_viz.reset_index(drop=True)

# define our 7 categories and assign a color to each one
category_colors = {
    'Cookies & Biscuits'      : '#E74C3C',   # red
    'Chips & Crisps'          : '#E67E22',   # orange
    'Chocolate & Candy'       : '#8E44AD',   # purple
    'Protein & Nutrition Bars': '#27AE60',   # green
    'Cereals & Breakfast'     : '#F1C40F',   # yellow
    'Nuts & Seeds'            : '#2E86C1',   # blue
    'Dairy Snacks'            : '#17A589',   # teal
}

print("Data ready for visualization")
print(f"Products available for charting: {len(df_viz):,}")
print(f"\nCategories and their colors:\n")
for category, color in category_colors.items():
    print(f"  {category:<30} {color}")
```

```
Data ready for visualization!
Products available for charting: 30,329

Categories and their colors:

  Cookies & Biscuits             #E74C3C
  Chips & Crisps                 #E67E22
  Chocolate & Candy              #8E44AD
  Protein & Nutrition Bars       #27AE60
  Cereals & Breakfast            #F1C40F
  Nuts & Seeds                   #2E86C1
  Dairy Snacks                   #17A589
```

SUGAR VS PROTEIN SCATTER PLOT

```
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

# set the size of the chart — width 14 height 8 inches
fig, ax = plt.subplots(figsize=(14, 8))

# loop through each category and plot its products as dots
for category, color in category_colors.items():
    # filter to only products in this category
    subset = df_viz[df_viz['primary_category'] == category]
    ax.scatter(
        subset['sugars_100g'],    # x axis — sugar content
        subset['proteins_100g'],  # y axis — protein content
        c=color,                  # color of the dots
        label=category,           # label for the legend
        alpha=0.4,                # transparency of dots — 0 invisible 1 solid
        s=20,                     # size of each dot
        edgecolors='none'         # no border around dots
    )

# add a shaded rectangle to highlight the Blue Ocean quadrant
# this is the key visual — high protein low sugar area
ax.axhspan(20, 100, xmin=0, xmax=0.15, alpha=0.1, color='green')
ax.text(2, 85, 'BLUE OCEAN\n(High Protein\nLow Sugar)',
        fontsize=10, color='green', fontweight='bold')

# add reference lines to divide the chart into quadrants
ax.axhline(y=20, color='gray', linestyle='--', linewidth=0.8, alpha=0.5)  # horizontal line at protein 20g
ax.axvline(x=20, color='gray', linestyle='--', linewidth=0.8, alpha=0.5)  # vertical line at sugar 20g

# add chart title and axis labels
ax.set_title('Sugar vs Protein — Where Is The Blue Ocean In The Snack Market?',
             fontsize=14, fontweight='bold', pad=15)
ax.set_xlabel('Sugar per 100g (g)', fontsize=12)
ax.set_ylabel('Protein per 100g (g)', fontsize=12)

# set the axis limits so the chart is not too stretched
ax.set_xlim(0, 100)
ax.set_ylim(0, 100)

# add the legend so we know which color is which category
ax.legend(title='Category', bbox_to_anchor=(1.01, 1), loc='upper left', fontsize=9)

# add a light grid so it is easier to read values
```

```
ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig('scatter_plot.png', dpi=150, bbox_inches='tight')  # save chart as image
plt.show()

print("Scatter plot saved as scatter_plot.png!")
```

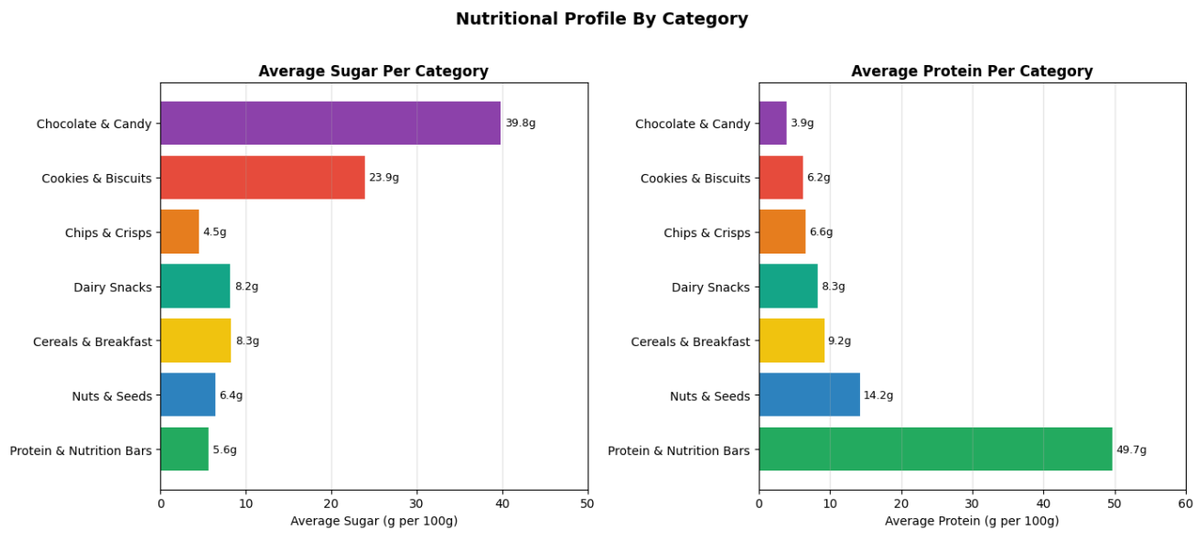



```
Scatter plot saved as scatter_plot.png!
```

AVERAGE NUTRITION BY CATEGORY

```
# calculate the average sugar and protein for each category

# this gives us one number per category instead of thousands of dots
category_avg = df_viz.groupby('primary_category').agg(
    avg_sugar   = ('sugars_100g',   'mean'),  # average sugar per category
    avg_protein = ('proteins_100g', 'mean')   # average protein per category
).round(1).reset_index()

# sort by average protein from highest to lowest
category_avg = category_avg.sort_values('avg_protein', ascending=False)

# create a figure with two charts side by side
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

# Chart 1 - Average Sugar Per Category
colors_sugar = [category_colors[cat] for cat in category_avg['primary_category']]
bars1 = ax1.barh(
    category_avg['primary_category'],  # categories on y axis
    category_avg['avg_sugar'],         # average sugar on x axis
    color=colors_sugar,                # color each bar by category
    edgecolor='none'                   # no border around bars
)

# add the exact number at the end of each bar
for bar, value in zip(bars1, category_avg['avg_sugar']):
    ax1.text(
        bar.get_width() + 0.5,   # position slightly after the bar ends
        bar.get_y() + bar.get_height() / 2,  # center vertically on bar
        f'{value}g',             # the number to display
        va='center', fontsize=9  # vertically centered small font
    )

ax1.set_title('Average Sugar Per Category', fontsize=12, fontweight='bold')
ax1.set_xlabel('Average Sugar (g per 100g)', fontsize=10)
ax1.set_xlim(0, 50)  # set x axis limit
ax1.grid(True, alpha=0.3, axis='x')  # light vertical grid lines

# Chart 2 - Average Protein Per Category
colors_protein = [category_colors[cat] for cat in category_avg['primary_category']]
bars2 = ax2.barh(
    category_avg['primary_category'],  # categories on y axis
    category_avg['avg_protein'],       # average protein on x axis
    color=colors_protein,              # color each bar by category
    edgecolor='none'                   # no border around bars
)

# add the exact number at the end of each bar
for bar, value in zip(bars2, category_avg['avg_protein']):
    ax2.text(
        bar.get_width() + 0.5,   # position slightly after the bar ends
        bar.get_y() + bar.get_height() / 2,  # center vertically on bar
```

```
        f'{value}g',             # the number to display
        va='center', fontsize=9  # vertically centered small font
    )

ax2.set_title('Average Protein Per Category', fontsize=12, fontweight='bold')
ax2.set_xlabel('Average Protein (g per 100g)', fontsize=10)
ax2.set_xlim(0, 60)  # set x axis limit
ax2.grid(True, alpha=0.3, axis='x')  # light vertical grid lines

plt.suptitle('Nutritional Profile By Category', fontsize=14,
             fontweight='bold', y=1.02)
plt.tight_layout()
plt.savefig('nutrition_by_category.png', dpi=150, bbox_inches='tight')
plt.show()

print("Chart saved as nutrition_by_category.png!")
print("\nAverage nutrition per category:\n")
print(category_avg.to_string(index=False))
```

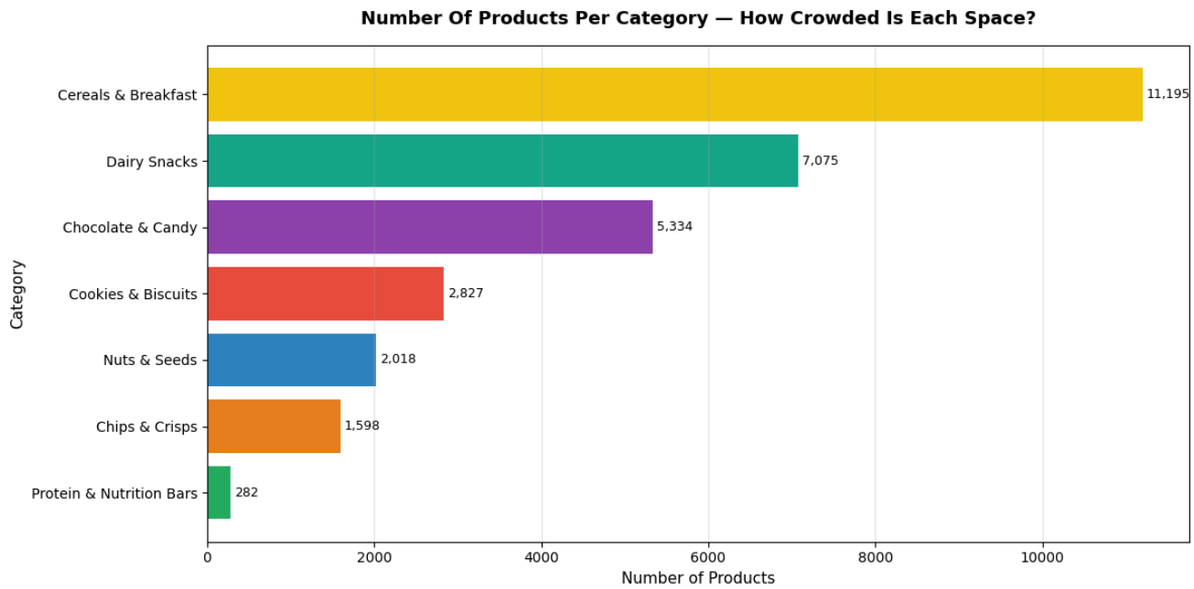



```
Chart saved as nutrition_by_category.png!

Average nutrition per category:

        primary_category  avg_sugar  avg_protein
Protein & Nutrition Bars        5.6         49.7
            Nuts & Seeds        6.4         14.2
     Cereals & Breakfast        8.3          9.2
            Dairy Snacks        8.2          8.3
          Chips & Crisps        4.5          6.6
       Cookies & Biscuits       23.9          6.2
        Chocolate & Candy       39.8          3.9
```

PRODUCT COUNT BY CATEGORY

```
# count how many products exist in each category
# this shows how crowded or empty each category is
category_counts = df_viz['primary_category'].value_counts().reset_index()
category_counts.columns = ['primary_category', 'count']

# sort by count from highest to lowest
category_counts = category_counts.sort_values('count', ascending=True)

# create the chart
fig, ax = plt.subplots(figsize=(12, 6))

# assign colors to each bar based on category
colors = [category_colors[cat] for cat in category_counts['primary_category']]

# draw horizontal bar chart
bars = ax.barh(
    category_counts['primary_category'],  # categories on y axis
    category_counts['count'],             # product count on x axis
    color=colors,                         # color each bar by category
    edgecolor='none'                      # no border around bars
)

# add the exact count at the end of each bar
for bar, value in zip(bars, category_counts['count']):
    ax.text(
        bar.get_width() + 50,            # position slightly after bar ends
        bar.get_y() + bar.get_height() / 2,  # center vertically on bar
        f'{value:,}',                    # format number with comma separator
        va='center', fontsize=9          # vertically centered small font
    )

# add chart title and axis labels
ax.set_title('Number Of Products Per Category — How Crowded Is Each Space?',
             fontsize=13, fontweight='bold', pad=15)
ax.set_xlabel('Number of Products', fontsize=11)
ax.set_ylabel('Category', fontsize=11)
```

```
# add a light grid so it is easier to read values
ax.grid(True, alpha=0.3, axis='x')

plt.tight_layout()
plt.savefig('product_count_by_category.png', dpi=150, bbox_inches='tight')
plt.show()

print("Chart saved as product_count_by_category.png!")
print("\nProduct count per category:\n")
for _, row in category_counts.iterrows():
    print(f"  {row['primary_category']:<30} {row['count']:>6,} products")
```




```
Chart saved as product_count_by_category.png!

Product count per category:

  Protein & Nutrition Bars          282 products
  Chips & Crisps                  1,598 products
  Nuts & Seeds                    2,018 products
  Cookies & Biscuits              2,827 products
  Chocolate & Candy               5,334 products
  Dairy Snacks                    7,075 products
  Cereals & Breakfast            11,195 products
```

THE RECOMMENDATION - STORY 4 (CELL 21)

```
# filter to only Protein & Nutrition Bars
# this is the category we identified as the Blue Ocean
protein_bars = df_viz[df_viz['primary_category'] == 'Protein & Nutrition Bars']

# calculate the average protein and sugar for this category
avg_protein = round(protein_bars['proteins_100g'].mean(), 1)
avg_sugar   = round(protein_bars['sugars_100g'].mean(), 1)

# median is the middle value — less affected by extreme outliers
median_protein = round(protein_bars['proteins_100g'].median(), 1)
median_sugar   = round(protein_bars['sugars_100g'].median(), 1)

# we use median as our target because it is realistic for
# a mainstream healthy snacking product not a supplement
target_protein = median_protein
target_sugar   = median_sugar

# print the heading
print("KEY INSIGHT & PRODUCT RECOMMENDATION\n")

# print the full recommendation in one block
print(f"""
Based on the analysis of {len(df_viz):,} snack products across 7 categories
the data reveals a clear market gap. The snack market is heavily saturated
in high sugar low protein categories like Chocolate & Candy (39.8g sugar, 3.9g protein)
and Cookies & Biscuits (23.9g sugar, 6.2g protein).

However the Protein & Nutrition Bars category is severely underserved
with only {len(protein_bars)} products in the entire dataset.

Based on the data the biggest market opportunity is in Protein & Nutrition Bars
specifically targeting products with {target_protein}g of protein and less than {target_sugar}g of sugar.

Supporting Data:
  - Average protein : {avg_protein}g    Average sugar : {avg_sugar}g
  - Median protein  : {median_protein}g    Median sugar  : {median_sugar}g
  - Total products in category : {len(protein_bars)}
  - Total products analyzed    : {len(df_viz):,}
""")
```

```
KEY INSIGHT & PRODUCT RECOMMENDATION


Based on the analysis of 30,329 snack products across 7 categories
the data reveals a clear market gap. The snack market is heavily saturated
in high sugar low protein categories like Chocolate & Candy (39.8g sugar, 3.9g protein)
and Cookies & Biscuits (23.9g sugar, 6.2g protein).

However the Protein & Nutrition Bars category is severely underserved
with only 282 products in the entire dataset.

Based on the data the biggest market opportunity is in Protein & Nutrition Bars
specifically targeting products with 58.1g of protein and less than 3.4g of sugar.

Supporting Data:
  - Average protein : 49.7g    Average sugar : 5.6g
  - Median protein  : 58.1g    Median sugar  : 3.4g
  - Total products in category : 282
  - Total products analyzed    : 30,329
```

BONUS STORY - THE HIDDEN GEM (CELLS 22-23)

FILTER HIGH PROTEIN PRODUCTS

```
# we define high protein as products with more than 20g of protein
# this is a common threshold used in sports nutrition
HIGH_PROTEIN_THRESHOLD = 20

# filter products that are high protein AND low sugar
# these are the products in our Blue Ocean quadrant
high_protein_products = df_viz[
    (df_viz['proteins_100g'] > HIGH_PROTEIN_THRESHOLD) &  # more than 20g protein
    (df_viz['sugars_100g']   < 20)                        # less than 20g sugar
].copy()

print(f"High protein threshold : above {HIGH_PROTEIN_THRESHOLD}g protein per 100g")
print(f"Low sugar threshold    : below 20g sugar per 100g")
print(f"Products in Blue Ocean : {len(high_protein_products):,}")
print(f"\nCategory breakdown of Blue Ocean products:\n")

# count how many Blue Ocean products exist in each category
blue_ocean_counts = high_protein_products['primary_category'].value_counts()
for category, count in blue_ocean_counts.items():
    pct = (count / len(high_protein_products)) * 100
    print(f"  {category:<30} {count:>5,}  ({pct:.1f}%)")
```

```
High protein threshold : above 20g protein per 100g
Low sugar threshold    : below 20g sugar per 100g
Products in Blue Ocean : 2,033

Category breakdown of Blue Ocean products:

  Dairy Snacks                     894  (44.0%)
  Nuts & Seeds                     709  (34.9%)
  Protein & Nutrition Bars         206  (10.1%)
  Cereals & Breakfast              124  (6.1%)
  Chocolate & Candy                 63  (3.1%)
  Cookies & Biscuits                22  (1.1%)
  Chips & Crisps                    15  (0.7%)
```

EXTRACT TOP PROTEIN INGREDIENTS

```
from collections import Counter
import re

# filter only high protein products that have ingredients listed
# we need the ingredients_text column to find protein sources
high_protein_with_ingredients = high_protein_products[
    high_protein_products['ingredients_text'].notna()  # only rows with ingredients
].copy()

print(f"High protein products with ingredients listed: {len(high_protein_with_ingredients):,}\n")

# these are the protein source keywords we are looking for
# we search for these words inside the ingredients text
protein_keywords = [
    'whey', 'peanut', 'almond', 'soy', 'casein',
    'egg', 'milk', 'pea protein', 'chickpea', 'lentil',
    'quinoa', 'hemp', 'walnut', 'cashew', 'pistachio'
]

# count how many times each protein keyword appears
# across all high protein product ingredients
keyword_counts = Counter()

for ingredients in high_protein_with_ingredients['ingredients_text']:
    # convert to lowercase so Whey and whey both match
    ingredients_lower = str(ingredients).lower()
    for keyword in protein_keywords:
        if keyword in ingredients_lower:   # check if keyword exists in ingredients
            keyword_counts[keyword] += 1   # add 1 to its count

# print the top 10 most common protein sources
print("Top protein sources found in high protein products:\n")
for rank, (ingredient, count) in enumerate(keyword_counts.most_common(10), 1):
```

```
    pct = (count / len(high_protein_with_ingredients)) * 100
    bar = '#' * int(pct / 2)
    print(f"  {rank}. {ingredient:<20} found in {count:>4} products  ({pct:.1f}%)  {bar}")
```

```
High protein products with ingredients listed: 1,226

Top protein sources found in high protein products:

  1. milk                 found in  486 products  (39.6%)  ###################
  2. peanut               found in  285 products  (23.2%)  ###########
  3. soy                  found in  178 products  (14.5%)  #######
  4. almond               found in   98 products  (8.0%)  ###
  5. whey                 found in   79 products  (6.4%)  ###
  6. pistachio            found in   33 products  (2.7%)  #
  7. pea protein          found in   27 products  (2.2%)  #
  8. lentil               found in   26 products  (2.1%)  #
  9. casein               found in   24 products  (2.0%)
  10. cashew               found in   24 products  (2.0%)
```

CANDIDATE'S CHOICE - HEALTH SCORE ANALYSIS

CALCULATE HEALTH SCORE

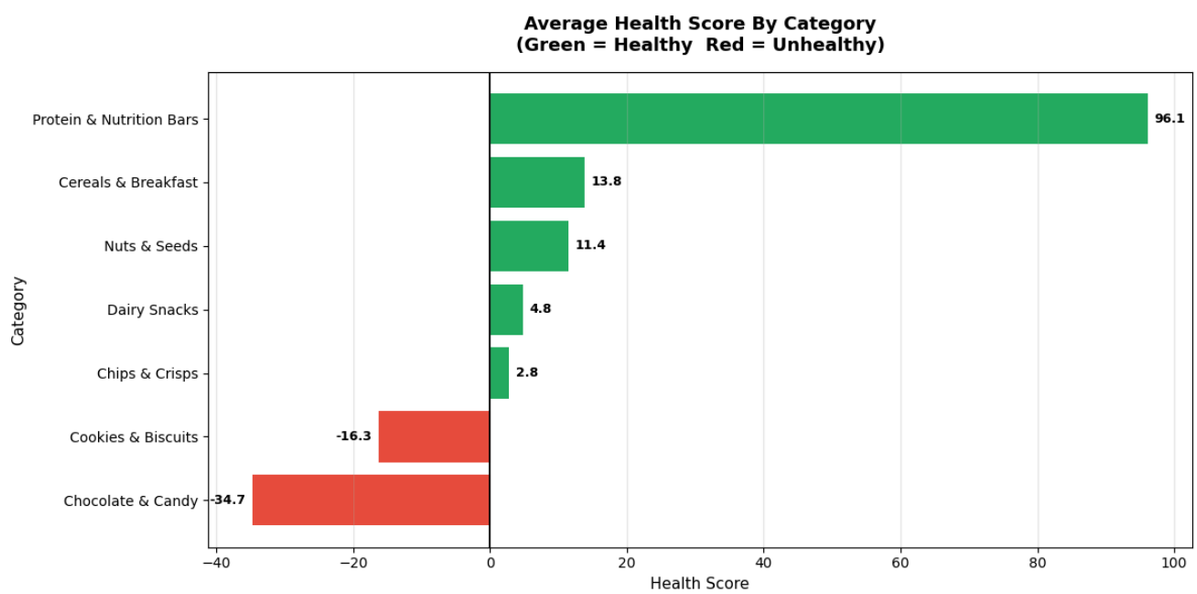
```
# this combines all nutrients into one easy to understand number
# the formula rewards good nutrients and penalizes bad ones
# Health Score = (protein x 2) + (fiber x 1.5) - (sugar x 1) - (fat x 0.5)

df_viz['health_score'] = (
    (df_viz['proteins_100g'] * 2.0) +         # reward protein heavily
    (df_viz['fiber_100g'].fillna(0) * 1.5) +  # reward fiber (fill NaN with 0)
    (df_viz['sugars_100g'] * -1.0) +           # penalize sugar
    (df_viz['fat_100g'] * -0.5)                # penalize fat lightly
).round(1)

print("Health scores calculated!\n")
print(f"Highest possible score : very positive number = very healthy")
print(f"Lowest possible score  : very negative number = very unhealthy")
print(f"\nOverall health score summary:\n")
print(f"  Average health score : {df_viz['health_score'].mean().round(1)}")
print(f"  Highest score        : {df_viz['health_score'].max()}")
print(f"  Lowest score         : {df_viz['health_score'].min()}")
print(f"\nAverage health score by category:\n")

# calculate average health score per category
# sort from highest to lowest so best categories appear first
health_by_category = (
    df_viz.groupby('primary_category')['health_score']
    .mean()
    .round(1)
    .sort_values(ascending=False)
)

for category, score in health_by_category.items():
    print(f"  {category:<30} {score:>6}")
```

```
Health scores calculated!

Highest possible score : very positive number = very healthy
Lowest possible score  : very negative number = very unhealthy

Overall health score summary:

  Average health score : 0.4
  Highest score        : 273.6
  Lowest score         : -100.0

Average health score by category:

  Protein & Nutrition Bars         96.1
  Cereals & Breakfast              13.8
  Nuts & Seeds                     11.4
  Dairy Snacks                      4.8
  Chips & Crisps                    2.8
  Cookies & Biscuits              -16.3
  Chocolate & Candy               -34.7
```

VISUALIZE HEALTH SCORE BY CATEGORY

```
# sort categories by health score from lowest to highest
# ascending=True puts the least healthy at the bottom
health_by_category_sorted = health_by_category.sort_values(ascending=True)

# create the chart
fig, ax = plt.subplots(figsize=(12, 6))

# assign colors based on score
# positive scores get green negative scores get red
bar_colors = [
    '#27AE60' if score > 0 else '#E74C3C'
    for score in health_by_category_sorted.values
]

# draw horizontal bar chart
bars = ax.barh(
    health_by_category_sorted.index,   # categories on y axis
    health_by_category_sorted.values,  # health scores on x axis
```

```
    color=bar_colors,                  # green or red based on score
    edgecolor='none'                   # no border around bars
)

# add the exact score at the end of each bar
for bar, value in zip(bars, health_by_category_sorted.values):
    x_position = value + 1 if value >= 0 else value - 1  # position label correctly
    alignment  = 'left' if value >= 0 else 'right'        # align label correctly
    ax.text(
        x_position,
        bar.get_y() + bar.get_height() / 2,  # center vertically on bar
        f'{value}',                           # score to display
        va='center',
        ha=alignment,
        fontsize=9,
        fontweight='bold'
    )

# add a vertical line at zero to separate healthy from unhealthy
ax.axvline(x=0, color='black', linewidth=1.2)

# add chart title and axis labels
ax.set_title('Average Health Score By Category\n(Green = Healthy  Red = Unhealthy)',
             fontsize=13, fontweight='bold', pad=15)
ax.set_xlabel('Health Score', fontsize=11)
ax.set_ylabel('Category', fontsize=11)

# add a light grid
ax.grid(True, alpha=0.3, axis='x')

plt.tight_layout()
plt.savefig('health_score_by_category.png', dpi=150, bbox_inches='tight')
plt.show()

print("Chart saved as health_score_by_category.png!")
```




```
Chart saved as health_score_by_category.png!
```

SAVE FINAL CSV FOR STREAMLIT

```
# at this point df_viz has all the columns we need for the dashboard
# save this as the final dataset that Streamlit will use

# check what columns we have in df_viz
print("Columns in final dataset:\n")
for col in df_viz.columns:
    print(f"  {col}")

# save the final dataset
df_viz.to_csv('food_facts_final.csv', index=False)

print(f"\nFinal dataset saved as food_facts_final.csv")
print(f"Total rows saved : {len(df_viz):,}")
print(f"Total columns    : {df_viz.shape[1]}")
```

```
Columns in final dataset:

  product_name
  categories_tags
  sugars_100g
  proteins_100g
  fat_100g
  fiber_100g
  energy_100g
  ingredients_text
  categories_clean
  primary_category
  health_score
```

```
Final dataset saved as food_facts_final.csv
Total rows saved : 30,329
Total columns    : 11
```